

Experimental comparison of tabular Q-learning performance on deterministic vs randomly generated environments

1. Setup

This experiment evaluates how tabular Q-learning behaves under different environmental conditions and hyperparameter settings.

Two environment types were tested:

- deterministic (stable) map generation
- randomly generated map generation

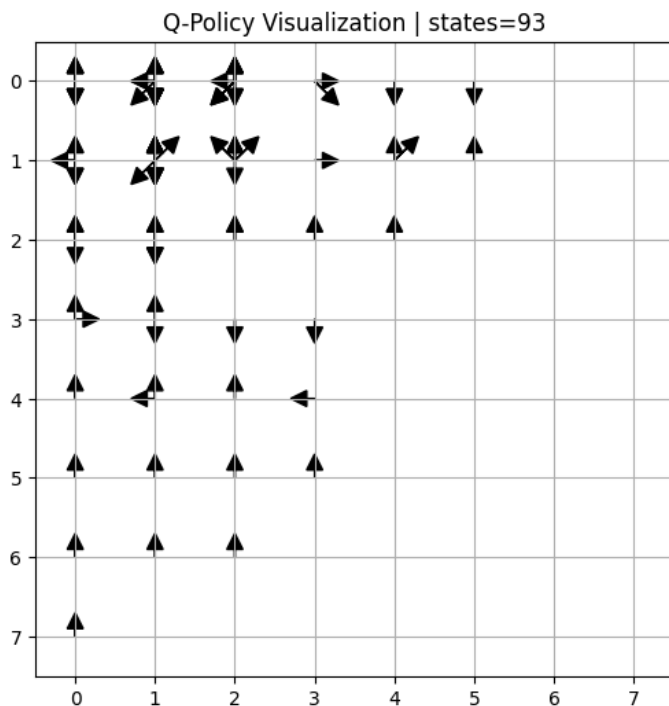
Default hyperparameters:

- γ (gamma): 0.99
- α (learning rate): 0.1
- ϵ (epsilon): 0.1

The agent uses:

- Q-table with standard Bellman update
 - ϵ -greedy exploration strategy
 - argmax action selection during exploitation
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2. Results on stable environment



Observations:

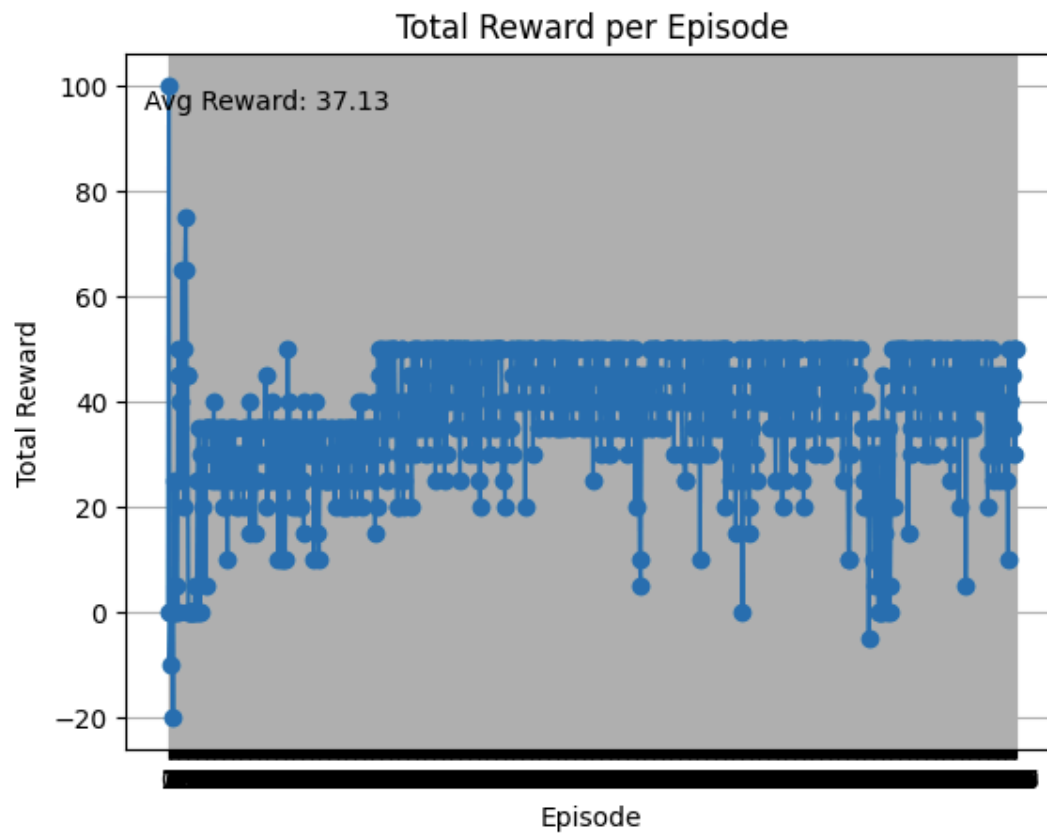
- The agent converges to a narrow subset of states and actions
- Exploration is extremely limited due to absence of curiosity mechanism
- Behavior stabilizes quickly but remains spatially concentrated near the starting region
- Action distribution shows noticeable variance despite convergence in Q-values

Key insight:

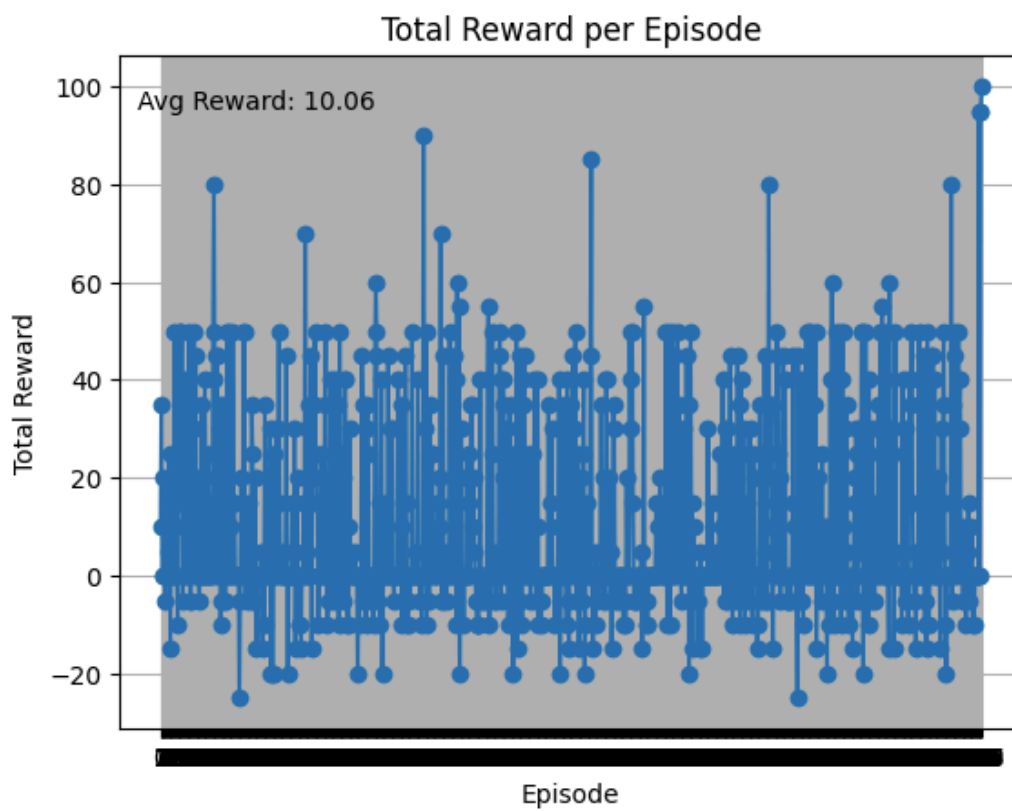
Even with convergence, the policy does not guarantee exploration of the state space, indicating structural limitation of ϵ -greedy alone in this setup.

Total reward comparison

Stable map:



Random map:

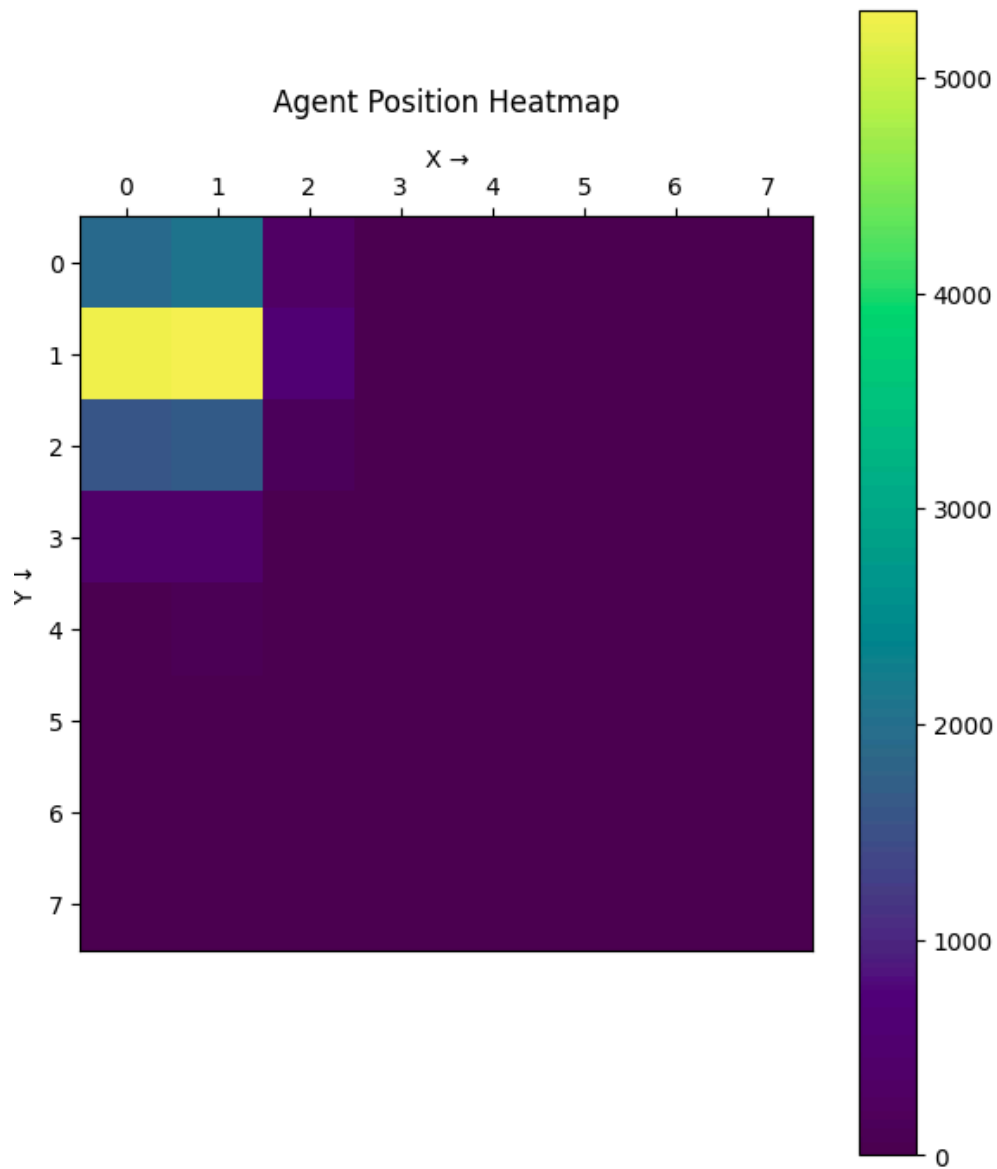


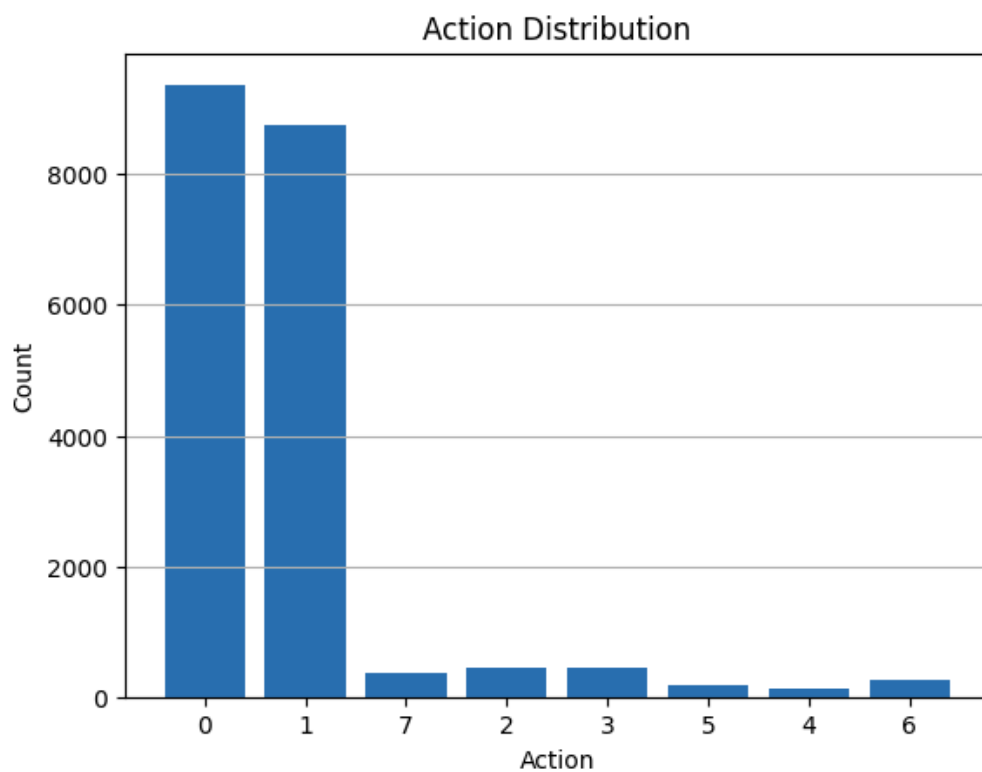
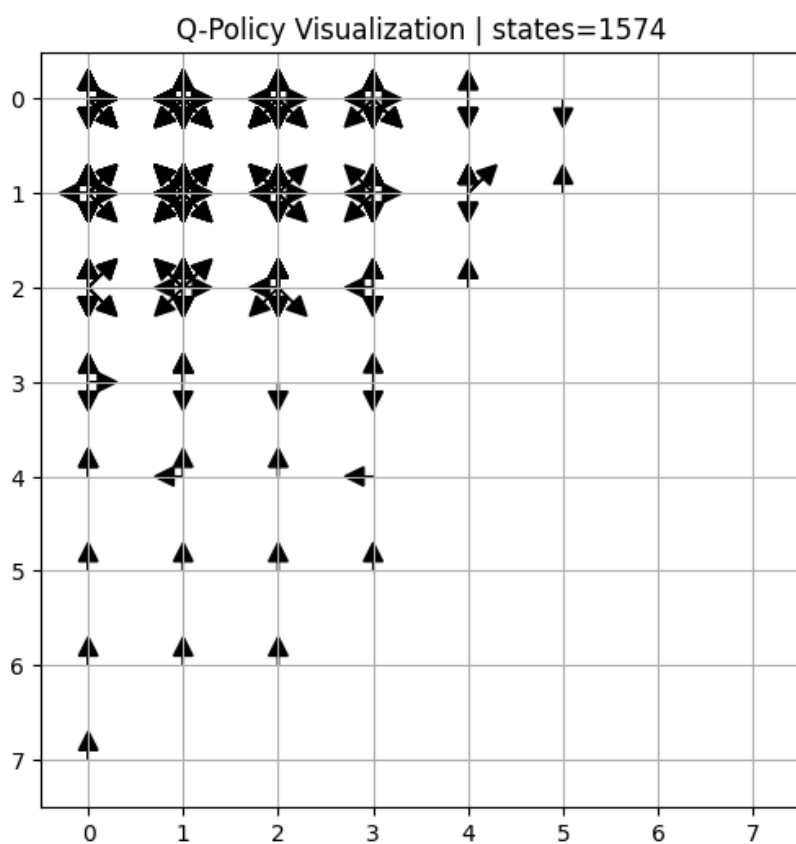
Observations:

- Significant drop in performance under stochastic transitions
- Higher variance in episode outcomes
 - more frequent negative returns
 - but also occasional high-reward spikes
- Indicates instability in policy generalization under non-deterministic structure

Additional observations (random vs stable)

Despite large differences in reward distribution, internal representations remain surprisingly similar:





Note:

Q-values appear denser in the combined dataset (2000 episodes total: stable + random), which affects visual interpretation but not underlying behavioral differences.

Unexpected behavior:

On the random map, the agent predominantly selects only actions:

- 0 (up)
- 1 (down)

Interpretation:

This behavior is not fully understood yet.

Possible hypotheses:

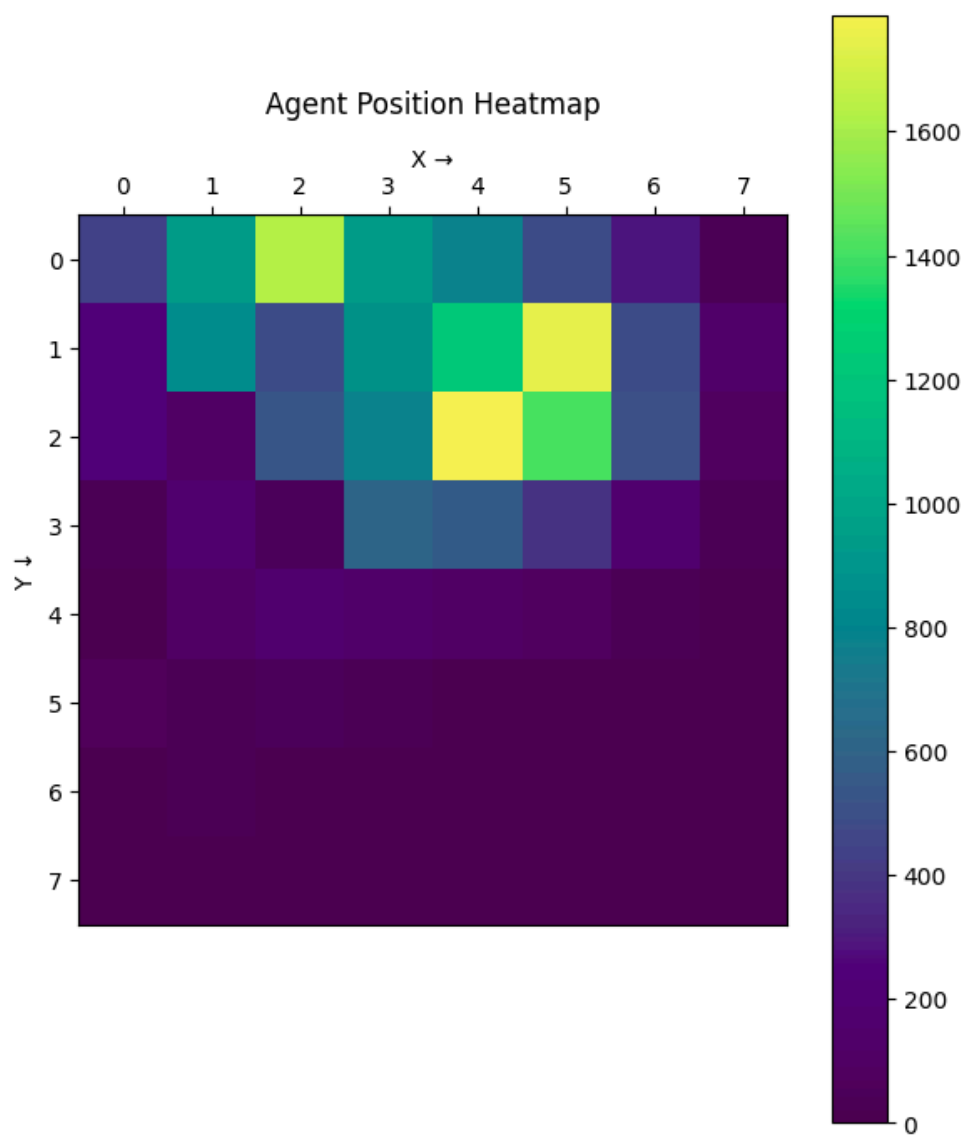
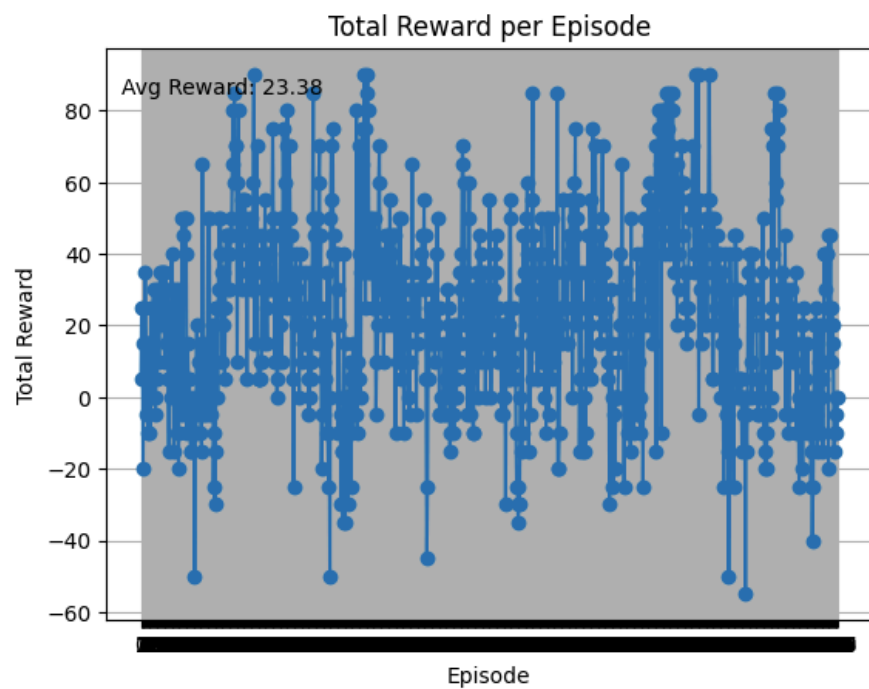
- collapse of policy toward locally stable actions
 - bias introduced by early Q-value updates
 - insufficient exploration pressure in stochastic environments
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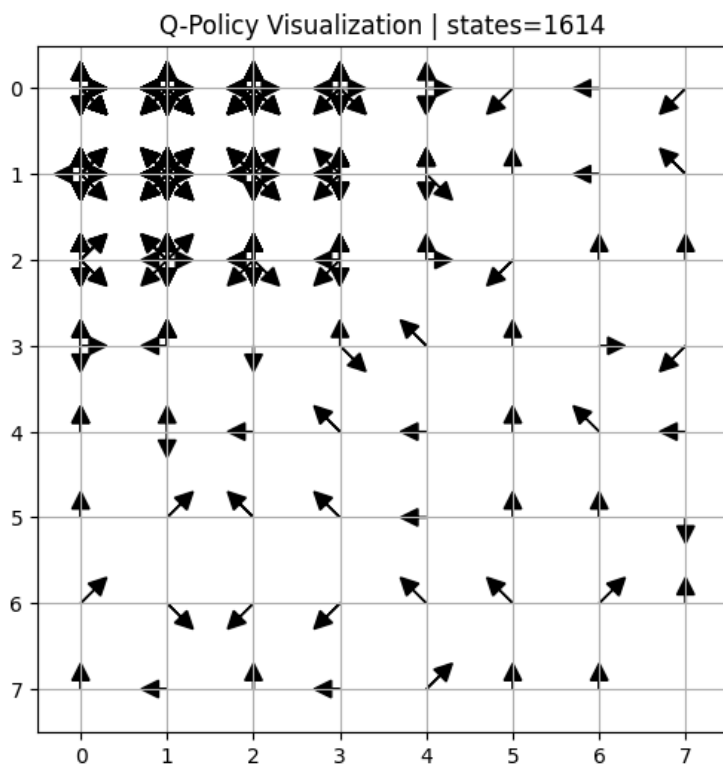
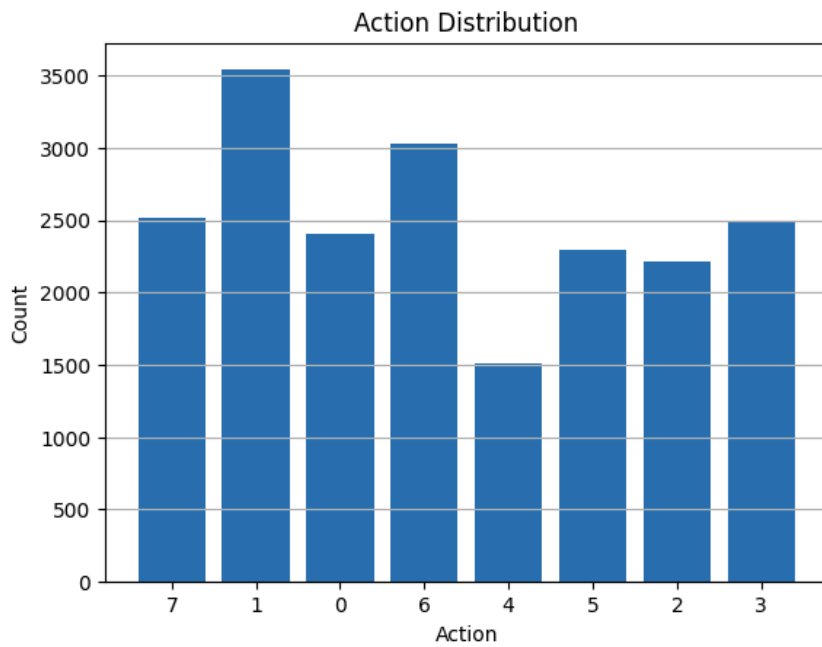
3. Experiment: increased learning rate ($\alpha = 0.5$)

New configuration:

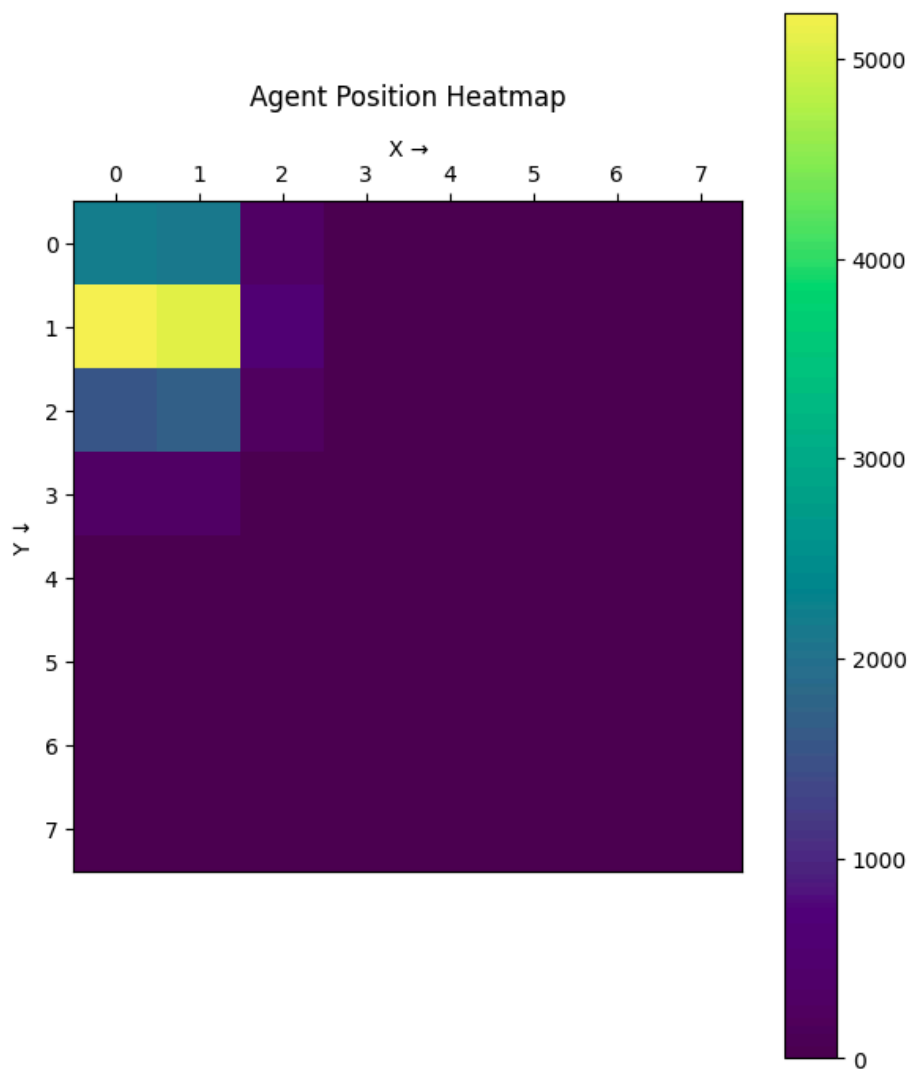
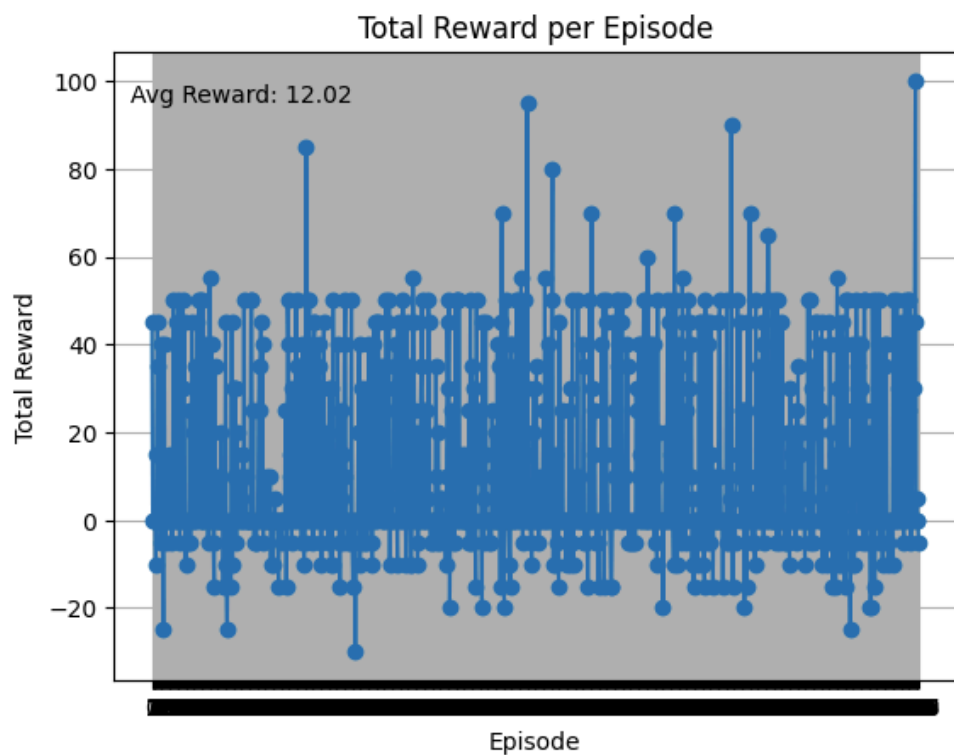
- γ (gamma): 0.99
 - α (learning rate): 0.5
 - ϵ (epsilon): 0.1
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Stable environment:





Random environment:



4. Discussion

The results indicate that:

- Lower learning rate ($\alpha = 0.1$) provides better overall stability and performance
- Higher learning rate ($\alpha = 0.5$) increases volatility and degrades performance in both environments
- Contrary to expectation, increased learning rate does not improve adaptability in stochastic environments

Interpretation:

Instead of improving responsiveness, higher α appears to:

- amplify noise in Q-value updates
- destabilize long-term value estimates
- reduce policy consistency under randomness

5. Final conclusion

Best-performing configuration in this setup:

- $\gamma = 0.99$
- $\alpha = 0.1$
- $\epsilon = 0.1$

Summary:

- agent maintains forward-looking behavior
- updates Q-values conservatively
- shows better stability under both deterministic and randomly generated maps

Open question:

Increasing learning rate was expected to improve adaptability in random environments, but experimental results suggest the opposite.

The underlying cause is still unclear and requires further investigation.